

Project 4 Guidelines

This project focuses on Tableau dashboards and story points. Select a dataset that contains both numerical and categorical variables. Using these variable types, you will create a variety of charts, including histograms, scatter plots, frequency tables, and pie charts.

For this assignment, you will need to find your own dataset. No data set will be provided. You may use any dataset that meets the requirements below, including those available on my project data page: <https://pengdsci.github.io/datasets/>.

4.1 Four Steps to Build an Effective Single-page Dashboard

Step 1: Draft your data story.

Begin by exploring the interconnected information in your data. Identify the key insights and shape them into a clear, logical narrative.

Step 2: Create interconnected, interactive charts.

Build individual charts that each reflect a piece of your story. Ensure the charts are interactive and connected to one another.

Step 3: Assemble the dashboard.

Combine your charts into a single dashboard layout. Add controls and filters based on relevant variables (such as time periods or categorical dimensions) to let viewers explore the story themselves.

Step 4: Apply visual design principles.

Review both the individual charts and the final dashboard. Make sure they follow data visualization's best practices for color, layout, labeling, and clarity.

4.1 The Minimum Requirements of Individual Charts

- **Numerical variables:** Choose a dataset containing at least two numerical variables, allowing you to construct a histogram and scatter plots.
- **Categorical variables:** The dataset must include at least three categorical variables so that you can create pie charts, bar plots, and similar visualizations.

- **Reactive filter:** Implement at least one filter that updates all charts in the dashboard (reactive dashboard).
- **Labels and titles:** Provide descriptive titles and axis labels for individual charts, as well as for the dashboard as a whole.
- **Layout and margins:** Adjust the margins of each chart and ensure the dashboard is tidy and informative.
- **Storytelling:** Develop a clear story point and include informative captions to support your narrative.
- **Optional – maps:** If your dataset contains geographic information, you may create maps, including animated maps.
- **Optional – advanced features:** You are encouraged to use advanced techniques such as derived variables and parameters to enhance your charts.

4.2 Submission Requirements

- Publish your dashboard and story point on Tableau Public Server.
- Write a report using an R Markdown (RMD) document that embeds both the Tableau dashboard and story point on the same web page.
- Publish the report either on RPubS or in your GitHub repository.
- **Optional:** Update your online CV with a link to this assignment.
- Submit the URL of your HTML report generated from the RMD file. Ensure that your RMD source code is downloadable from the HTML report.